Assignment 1 (September 16, 2022)

Please provide calculations and explanations where it is required and submit your response as a **single PDF** file by no later than **5:00 PM next Monday**.

1. Chromium, Cr, has the following isotopic masses and fractional abundances (2 pts):

Mass number	**Isotopic mass (amu)**	**Fractional abundance**
50	49.9461	0.0435
52	51.9405	0.8379
53	52.9407	0.0950
54	53.9389	0.0236

What is the atomic mass of chromium? Provide calculations with your answer.

2. Balance the following chemical equations (1 pt for each):

a. $H_3PO_3 \rightarrow H_3PO_4 + PH_3$

b. $Fe_2(SO_4)_3 + NH_3 + H_2O \rightarrow Fe(OH)_3 + (NH_4)_2SO_4$

3. Formaldehyde, CH_2O, is a toxic gas with a pungent odor. Calculate the mass percentages of the elements in formaldehyde? (2 pts)

4. In a process for producing acetic acid, oxygen gas is bubbled into acetaldehyde, CH_3CHO, containing manganese (II) acetate (catalyst) under pressure at 60°C.

$2CH_3CHO(l) + O_2(g) \rightarrow 2HC_2H_3O_2(l)$

In a laboratory test of this reaction, 20.0g CH_2CHO and 10.0g O_2 were put into a reaction vessel.

a. Which chemical is the limiting reagent here? Provide calculations with your answer (2 pts).

b. How many grams of acetic acid can be produced by this reaction from these amounts of reactants (2 pts)?

c. If the actual yield of acetic acid from this reaction is 23.8g, then what is the percentage yield of acetic acid (2 pts)?

d. How many grams of excess reactant remain after the reaction is complete (2 pts)?

5. Write the net ionic equation of the following equation (2 pts):

$ZnS(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2S(g)$

6. Calculate oxidation number of Cl in ClO_3^- (2 pts).

7. A sample of $NaNO_3$ weighing 0.38g is placed in a 50.0mL volumetric flask. The flask is then filled with water to the mark on the neck, dissolving the solid. What is the molarity of the resulting solution? Provide calculations with your answer (2 pts).